Prevalence of Homologous Recombination Pathway Gene Mutations in Melanoma: Rationale for a New Targeted Therapeutic Approach

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Homologous recombination DNA damage repair (HR-DDR) deficient patients with various solid tumors have been treated with PARP inhibitors. However, the clinical characteristics of patients with melanoma who have HR-DDR gene mutations and the consequences of PARP inhibition are poorly understood. We compared the commercially available next-generation sequencing data from 84 patients with melanomas from our institution with a dataset of 1,986 patients as well as 1,088 patients profiled in cBioportal. In total, 21.4% of patients had ≥1 functional HR-DDR mutation, most commonly involving *BRCA1*, *ARID1A*, *ATM*, *ATR*, and *FANCA*. Concurrent *NF1*, *BRAF*, and *NRAS* mutations were found in 39%, 39%, and 22% of cases, respectively. HR-DDR gene mutation was associated with high tumor mutational burden and clinical response to checkpoint blockade. A higher prevalence of HR-DDR mutations was observed in the datasets from Foundation Medicine (Cambridge, CA) and those from the Cancer Genome Atlas. Treatment of HR-DDR–mutated patient-derived xenograft models of melanoma with PARP inhibitor produced significant antitumor activity in vivo and was associated with increased apoptotic activity. RNA sequencing analysis of PARP inhibitor–treated tumors indicated alterations in the pathways involving extracellular matrix remodeling, cell adhesion, and cell-cycle progression. Melanomas with HR-DDR mutations represent a unique subset, which is more likely to benefit from checkpoint blockade and may be targeted with PARP inhibitor.

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INTRODUCTION

The development of checkpoint inhibitors and, for *V600 BRAF*-mutant melanoma, BRAF/MAPK/(extracellular signal–regulated kinase) kinase–targeted therapies has significantly prolonged the overall survival (OS) of patients with advanced melanoma (Chapman et al., 2011; Hauschild et al., 2012; Larkin et al., 2014; Robert et al., 2015). However, a majority of patients will ultimately experience disease progression and will need subsequent treatments. The identification of novel, actionable molecular targets for these patients would be highly desirable.

PARP inhibition is one of the latest molecular targeting strategies in cancer. PARP is directly involved in the prevention of DNA double-strand breaks and is required for proper homologous recombination DNA damage repair (HR-DDR) (Bryant et al., 2009). Accordingly, the deficiency or dysfunction of HR-DDR sensitizes cancer cells to PARP inhibitor (PARPi), resulting in chromosome instability, cell-cycle arrest, and apoptosis (Ashworth, 2008; Farmer et al., 2005). The loss or disruption of proteins necessary for HR-DDR is observed in many tumor types and may confer sensitivity to PARPis (Bell et al., 1999; Bullrich et al., 1999; Couch et al., 2005; Konstantinopoulos et al., 2010; Marsit et al., 2004; McCabe et al., 2006; Miller et al., 2002; Narayan et al., 2004; Turner et al., 2004; Vorechovsky et al., 1999; Walsh et al., 2011). Clinically, PARPis have shown promising clinical activity in patients with homologous recombination (HR)-deficient tumors (Mirza et al., 2016; Swisher et al., 2017), with a prolongation of progression-free survival and/or OS in various cancers with a *BRCA* mutation and/or HR deficiency, including ovarian, breast, pancreatic, and prostate cancers (de Bono et al., 2020; Golan et al.,

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Abbreviations: FMI, Foundation Medicine; HR, homologous recombination; HR-DDR, homologous recombination- DNA damage repair; MM, melanoma model; OS, overall survival; PARPi, PARP inhibitor; PDX, patient-derived xenograft; TMB, tumor mutational burden

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2019; Litton et al., 2018; Mirza et al., 2016; Robson et al., 2017; Swisher et al., 2017), and have obtained regulatory approval for some of these cancers.

In this study, we investigated the frequency of functional mutations in HR-DDR pathway genes in patients with advanced melanoma, along with their clinical and pathologic characteristics, and investigated the preclinical activity of PARPi in melanoma cell lines with HR-DDR pathway gene mutations.

RESULTS

Frequency of HR-DDR pathway gene mutations

Genomic analysis was conducted on samples from 84 patients with metastatic (i.e., stage III or IV) cutaneous melanoma. Tumors of 18 patients (21.4%) harbored an alteration in at least one of the HR genes categorized as pathogenic or likely pathogenic, which was known or considered to result in a change to the function of the protein likely to disrupt DNA repair. The most commonly altered genes were *BRCA1* and *ARID1A* (3.6% each), followed by *ATM, ATR,* and *FANCA* (2.4% each) (Table 1). Mutations in *ATRX, BRIP1, BAP1, CHEK2, BARD1, MRE11, FANCD2, ARID1B,* and *PALB2* (1.2%) were found in one patient each. Two patients had concurrent mutations in multiple HR-DDR pathway genes within the same melanoma sample: one with mutations in *ARID1A, BRCA1,* and *MRE11* and the other with mutations in *ARID1B* and *BRCA1*. Among patients with HR mutation(s), concurrent *NF1*, V600 *BRAF*, *NRAS*, and *KIT* mutations were found in seven patients (38.9%), seven patients (38.9%), four patients (22.2%), and one patient (5.6%), respectively. To confirm the frequency of HR deficiency in a larger cohort of patients with melanoma, next-generation sequencing data of 1,986 melanoma samples performed by Foundation Medicine (FMI) (Cambridge, CA) were evaluated. Analysis of the FMI cohort identified 33.5% of cases with HR-DDR alterations (Table 1), consistent with our findings. In this dataset, the most common mutated gene was *ARID1A* (5.0%), followed by *ATM* (4.0%), *BAP1* (3.1%), *ATRX* (2.8%), *BRCA2* (2.3%), *FANCG/L* (2%), *ATR* (1.6%), *BRCA1* (1.3%), *BRIP1* (1.1%), *NBN* (1.0%), and *FANCA* (1.0%). Interestingly, 15 of the mutations identified in HR-DDR genes have not been previously reported in the Catalogue of Somatic Mutations in Cancer database (https://cancer.sanger.ac.uk/cosmic and Supplementary Table S1). Analysis of the 1,088 profiled patients with melanoma in cBioportal also identified a high prevalence of HR-DDR pathway gene mutations (Table 1).

Patients and tumor characteristics

Among the 18 patients with HR-deficient melanoma, the median age was 68 years. A total of 14 patients (77.8%) were male, and 11 patients (73.3%) had a primary melanoma in the head and neck (Table 2). Among the 14 patients with HR deficiency with known status of ulceration and dermal mitotic rates in the primary tumor, 12 (85.7%) had non-ulcerated melanoma and 12 (85.7%) had a mitotic rate $\geq 1/mm^2$. Among 16 patients with known tumor mutational burden (TMB), TMB was high in 11 patients (68.8%) and low in 3 patients (18.8%). Among nine patients with HR deficiency who were treated with anti–PD-1 antibody (with or without anti-CTLA4 antibody), six (66.7%) had a clinical response, and only one patient (11.1%) experienced early disease progression within 3 months.

These clinical and pathologic characteristics were compared with those of 66 patients whose tumors did not harbor an HR-DDR pathway gene alteration. HR-DDR pathway alterations were significantly associated with thinner primary tumors, a higher proportion of head and neck primary melanoma, high TMB, and increased proportion of responders to anti–PD-1 antibody–based therapy (Table 2). Interestingly, there was a significantly higher proportion of thinner tumors (up to 2-mm thick) in the HR-DDR–mutated group ($P = 0.03$, Table 2). In addition, 45% of the patients without HR-DDR alteration had early disease progression within 3 months after starting anti–PD-1–based therapy compared with 11.1% in the HR-DDR–mutated group ($P = 0.03$). By univariate Cox regression analysis, increasing age and tumor thickness as well as response to PD-1 blockade were significantly predictive of OS, whereas there was no significant association between HR-DDR pathway gene mutational status and OS (Table 3). Multivariate analysis of the factors significant on univariate analysis identified PD-1 response as the most significant factor predictive of survival in this cohort ($P = 0.008$), followed by age ($P = 0.046$) and tumor thickness ($P = 0.048$).

Effects of PARP inhibition on melanoma cells with HR pathway mutation

Although the tissue-based analysis presented earlier determined a relatively high prevalence of HR-DDR pathway gene mutations in melanoma, whether the presence of such mutations confers sensitivity to PARPis in melanoma cells has not been clearly demonstrated. To this end, we screened our library of in-house patient-derived xenograft (PDX) models of melanoma and identified two lines for further analysis: MM425 (harboring mutations in *BRCA1* and *ARID1B*) and MM390 (harboring a mutation in the *CHD2* gene). The *CHD2* gene was of interest because it directly interacts with PARP1 (Luijsterburg et al., 2016) even though it was not included in the original list of HR-DDR pathway genes. Analysis of our institutional, FMI, and cBioportal datasets indicated a 1%, 1%, and 4% prevalence of *CHD2* mutations in melanoma, respectively.

Initially, a dose-finding study was conducted to determine the tolerable doses of two PARPis, niraparib and olaparib, in NSG mice. On the basis of this, a dose of 25 mg/kg (for niraparib; #S2741, Selleck Chemicals, Houston, TX) and a dose of 50 mg/kg (for Olaparib; #S1060, Selleck Chemicals) were selected for antitumor studies. Daily intraperitoneal injections of either olaparib or niraparib resulted in significant antitumor effects against both MM425X (Figure 1a) and MM390X (Figure 1b) cells. To determine the transcriptomic profiles altered after niraparib treatment, RNA sequencing was performed in MM425 tumors from three mice treated with niraparib compared with three mice treated with vehicle. Gene ontology analysis indicated alterations in the cell cycle, cell movement, integrin signaling, and collagen and matrix remodeling as well as in triglyceride metabolism and fatty acid and cholesterol biosynthesis pathways. Supervised analysis of the niraparib-treated samples versus that of

Table 1. Frequency of Functional HR-DDR Pathway Gene Mutations in Melanoma

Gene	CPMCRI Cohort (n = 84[1]), n (%)	Foundation Medicine Cohort (n = 1,986), %	cBioportal Cohort (9 studies; n = 1,088[2]), %
BRCA1	3 (3.6)	1.3	5
ARID1A	3 (3.6)	5	7
ATM	2 (2.4)	4	7
ATR	2 (2.4)	1.6	7
FANCA	2 (2.4)	1	4
ATRX	1 (1.2)	2.8	7
BRIP1	1 (1.2)	1.1	4
BAP1	1 (1.2)	3.1	2.3
CHEK2	1 (1.2)	0.7	2.5
BARD1	1 (1.2)	0.2	1.7
MRE11	1 (1.2)	<1	2.4
FANCD2	1 (1.2)	0.2	5
ARID1B	1 (1.2)	0.1	6
PALB2	1 (1.2)	0.5	4
NBN	0 (0)	1	2.1
BRCA2	0 (0)	2.3	8
FANCC	0 (0)	0.4	1.2
FANCE	0 (0)	0	1.7
FANCF	0 (0)	0	0.3
FANCG	0 (0)	2	1.5
FANCL	0 (0)	2	1.7
WRN	0 (0)	ND	5
BLM	0 (0)	0.2	3
RAD50	0 (0)	0.9	2.4
RAD51	0 (0)	0.2	0.8
RAD51B	0 (0)	0	1.3
RAD51C	0 (0)	0.3	1.3
Total	18[1] (21.4)	33.5	41[2]

Abbreviations: CPMCRI, California Pacific Medical Center Research Institute; HR, homologous recombination; HR-DDR, homologous recombination DNA damage repair; ND, not determined.

[1]A total of 18 of 84 patients harbored at least one HR pathway gene mutation.

[2]A total of 444 of 1,088 patients harbored at least one HR pathway gene mutation.

vehicle-treated samples identified 268 differentially expressed genes (Figure 1c). Of these, 28 genes were upregulated and 16 were downregulated at least 1.5-fold (Supplementary Table S2). Among the significantly differentially expressed genes were *AXL*, *COXL1*, *COL1A2*, *COL3A1*, *DPT*, and *LOXL1* (downregulated) as well as *ALDOC*, *ALDH3B2*, and *FADS2* (upregulated). The differential expression of several of these genes was confirmed by quantitative real-time reverse transcriptase–PCR analysis. Niraparib-induced modulation of gene expression was validated for all the genes listed earlier, whereas olaparib had a similar effect on expression levels of most of the genes analyzed (Figure 1d).

We then performed additional analyses in culture to further characterize the consequences of niraparib treatment in melanoma cell lines. We initially assessed the effects of niraparib treatment on melanoma cell viability. In vitro administration of niraparib at concentrations between 1 μM and 60 μM resulted in the suppression of the survival of both MM425X and MM390X cells (Figure 1e and f, respectively). In contrast, treatment of MM507X cells, which did not contain detectable HR mutations, resulted in minimal suppression of survival (Figure 1g). Treatment of MM425X and MM390X cells also resulted in the induction of apoptosis (Figure 1h and i). Because neither PDX line persists in the culture for an extended period of time, we also evaluated the effects of niraparib treatment on the established LOX human melanoma cell line, which harbors mutations in *ARID1A* and *BAP1* genes. Short-term (72-hour) niraparib treatment suppressed LOX cell viability (Figure 2a), whereas longer-term (8-day) exposure resulted in profound suppression of the colony formation capacity of LOX cells (Figure 2b and c). The growth suppression observed was accompanied by marked induction of apoptotic activity (Figure 2d) as well as

Table 2. Clinical and Pathologic Characteristics of Patients with HR Pathway Gene Mutations

Characteristics	HR Mutation (n = 18)	Non-HR Mutation (n = 66)	*P*-Value
Age, y, median (range)	68 (52–81)	65 (21–91)	0.21
Sex, n (%)			
Male	14 (77.8)	40 (60.6)	0.41
Female	4 (22.2)	26 (39.4)	
Primary melanoma site, n (%)			
Head and neck	11 (73.3)	15 (25.4)	0.001
Torso	3 (20)	14 (23.7)	
Extremities	1 (6.7)	30 (50.9)	
Melanoma thickness (mm), n (%)			
Up to 1	3 (21.4)	4 (7.3)	0.03
>1–2	5 (35.7)	15 (27.3)	
>2–4	1 (7.2)	25 (45.4)	
>4	5 (35.7)	11 (20)	
Median	1.6	2.6	0.46
Ulceration status, n (%)			
Present	2 (14.3)	25 (46.3)	0.06
Absent	12 (85.7)	29 (53.7)	
Mitotic rate (mitosis/mm^2), n (%)			
<1	2 (14.3)	2 (4.2)	0.57
≥1	12 (85.7)	46 (95.8)	
Median	6	3	0.41
Tumor mutation burden, n (%)			
Low	3 (18.8)	27 (43.5)	0.03
Intermediate	2 (12.5)	15 (24.2)	
High	11 (68.8)	20 (32.3)	
Median	10.8	65.8	0.02
Concurrent genetic mutation, n (%)			
NF1	7 (38.9)	7 (10.6)	0.03
NRAS (Q61, G12, G13)	4 (22.2)	23 (34.9)	0.57
BRAF (V600)	7 (38.9)	28 (42.4)	1.00
KIT	1 (5.6)	1 (1.5)	1.00
Median time to distant metastasis from initial diagnosis, mo (range)	45.0 (0.7–243)	25.0 (0–247)	0.81
Response to anti–PD-1 antibody inhibitors (±ipilimumab), n (%)			
Response	6 (66.7)	16 (40.0)	
Stable disease	2 (22.2)	6 (15.0)	
Progressive disease	1 (11.1)	18 (45.0)	0.03

Abbreviation: HR, homologous recombination.

Table 3. Univariate Cox Regression Analysis of Clinical Factors in Patients with Metastatic Melanoma: Time to Death from the Initial Diagnosis

Clinical Factor	n	Chi-Square	*P*-Value	Hazard Ratio	Comparative Factors
HR pathway gene mutation	82	0.84	0.36	0.64	—
Age	82	5.51	0.019	1.39	—
Gender	82	0.86	0.35	1.41	—
Primary site	73	0.70	0.40	0.74	—
Melanoma thickness	68	4.53	0.03	1.58	—
Ulceration	67	1.2	0.27	1.54	—
Mitotic rate	61	0.30	0.58	1.02	—
Tumor mutation burden	76	2.61	0.11	1.82	Low versus high + Intermediate
Concurrent genetic mutation					
BRAF	82	0.77	0.38	0.72	—
NRAS	82	0.38	0.54	0.79	—
NF1	82	1.02	0.31	1.55	—
Response to PD-1 antibody	39	9.33	0.002	7.13	PD versus PR + CR
	47	6.44	0.01	2.92	PD versus PR and/or CR and/or SD

Abbreviations: CR, complete response; HR, homologous recombination; PD, progressive disease; PR, partial response; SD, stable disease.

pronounced changes in the cell cycle, including reduced S phase and a G2/M block (Figure 2e). As described earlier, RNA sequencing analysis of niraparib-treated PDX samples showed inhibition of pathways regulating cell movement and matrix remodeling. Because MM425X and MM390X PDX lines did not exhibit an invasive behavior in Matrigel or vitronectin-coated transwell assays, we used LOX cells to interrogate the functional consequences of niraparib-induced changes on melanoma cell invasion. As shown in Figure 2f and g, both 3 μM and 6 μM niraparib treatment induced significant inhibition of cell invasion into Matrigel. Furthermore, TaqMan analyses of niraparib-treated LOX cells compared with that of vehicle-treated LOX cells showed a downregulation of the expression of the following genes known to regulate cell invasion and migration: *AXL*, *COL3A1*, *DPT,* and *LOXL1* (similar to those of MM425X, Figure 2h).

DISCUSSION

In this study, we show that HR-DDR pathway gene mutations occur frequently in cutaneous melanoma because 21.4% of melanomas harbored at least one HR-DDR pathway gene alteration in our cohort. Analysis of our cohort identified mutations in several genes that have not been previously reported in the Catalogue of Somatic Mutations in Cancer database either in melanoma or in any other malignancy. Our findings were corroborated by analyses of two larger cohorts, with a prevalence of 33.5% in the FMI cohort and a prevalence of 41% in cBioportal. In our cohort, the presence of HR-DDR pathway gene mutation was associated with a significantly higher proportion of thinner primary tumors, a higher proportion of head and neck primary tumors, higher TMB, and concurrent *NF1* mutation. Intriguingly, in a small number of patients treated with systemic therapy, the majority of patients with HR-DDR pathway gene mutations responded to anti–PD-1 antibody–based immunotherapy, including a significantly higher objective response rate to a single-agent anti–PD-1 antibody (66.7% vs. 40.0%). It is difficult to ascertain whether the presence of HR deficiency or TMB is a better predictor of response to checkpoint inhibition in advanced cutaneous melanoma at this time. It should be noted that the definition of a high TMB in our analysis is different from the more recent definition of high TMB (≥10 mutations per megabase) for which pembrolizumab treatment is indicated in other solid tumors. Clinical responses to BRAF inhibitors or cytotoxic chemotherapy could not be evaluated because few patients were treated with these agents. Although in our analysis there was no significant difference in OS between patients with HR deficiency and those without, the number of patients was too small to make a definite conclusion about the prognostic impact of these mutations. However, response to PD-1 blockade was significantly associated with prolonged survival and was the most significant factor in multivariate analysis.

To our knowledge, there have not been any other studies to characterize the HR-DDR mutant population in cutaneous melanoma. Heeke et al. (2018) conducted a pan-cancer analysis of 52,426 patients across multiple solid tumors undergoing next-generation sequencing using a different commercial platform in which melanoma was included. This study reported an 18.1% prevalence of HR-DDR mutations in melanoma. Differences between the two studies include differences in the testing platforms used, differences in the gene set used to define HR-DDR, as well as the specific focus on cutaneous melanoma in our study. Nevertheless, both analyses indicate that HR-DDR mutations are common events in melanoma.

Beyond the apparent positive correlation between the presence of HR-DDR pathway gene mutations and response to checkpoint inhibitor therapy, the therapeutic implications of these mutations in the management of advanced melanoma are not clear. The preclinical efficacy evaluation in this study confirmed our hypothesis that mutations within HR-DDR pathways are associated with response to two different PARPi in two distinct PDX models with mutations in different HR-DDR pathway genes. These studies also identify

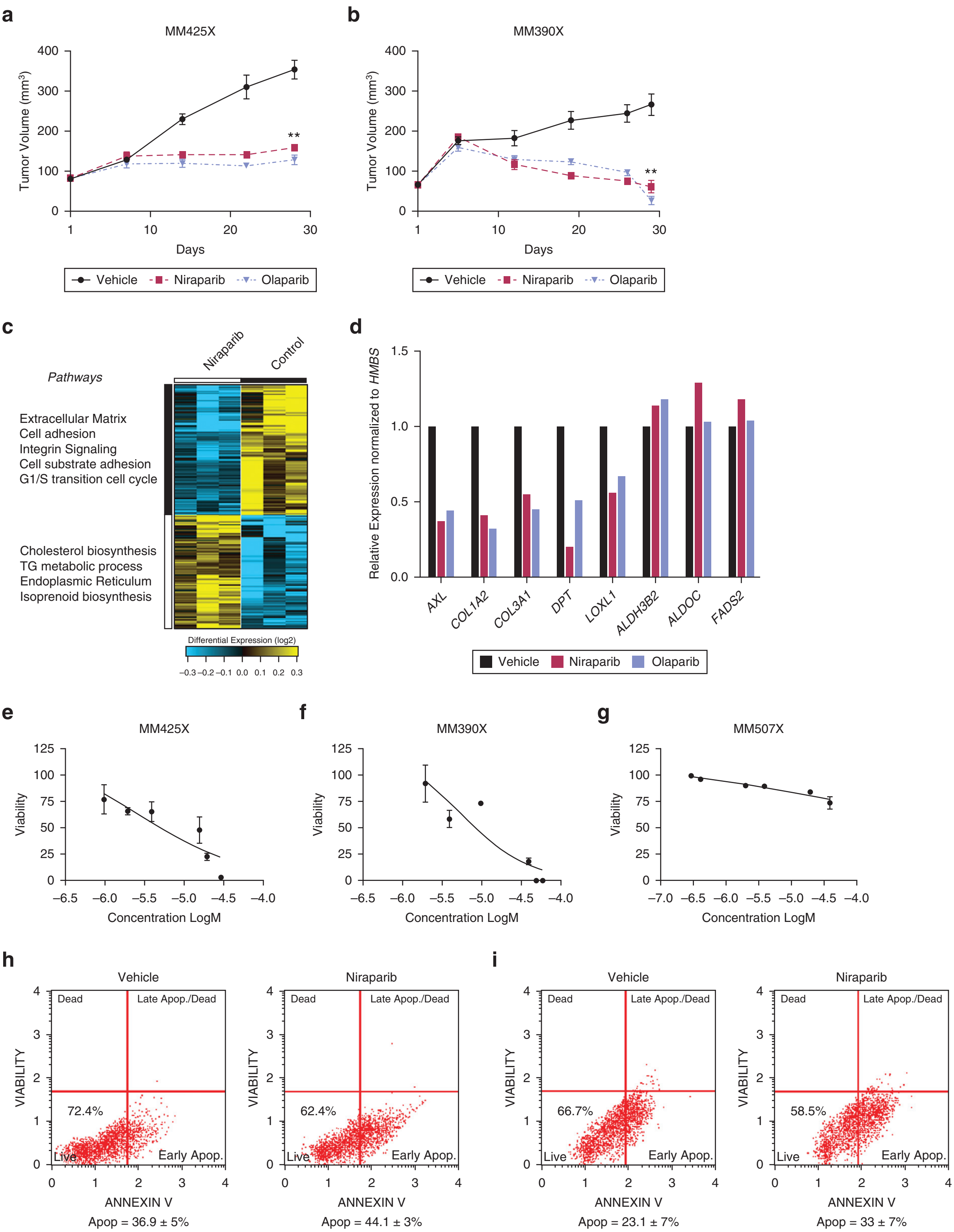

Figure 1. PARPis exert a potent antitumor effect against melanoma PDXs in vivo. (**a**) MM425X- and (**b**) MM390X-bearing NSG mice were treated with niraparib (25 mg/kg) or olaparib (50 mg/kg) for 30 days. $^{**}P < 0.0001$ two-way ANOVA. (**c**) Heatmap of the RNA-Seq analysis of niraparib-treated samples versus that of vehicle-treated MM425X samples (three mice per group) representing the top 268 up and downregulated genes with key modulated pathways (left). (**d**) Relative expression of *AXL, COL1A2, COL3A1, DPT, LOXL1, ALDH3B2, ALDOC,* and *FADS2* genes by qRT-PCR in MM425X tumors from mice treated with vehicle, olaparib, or niraparib and normalized to the human *HMBS* gene. Cell survival analysis of (**e**) MM425X, (**f**) MM390X, and (**g**) MM507X treated with niraparib (1–60 μM; treated for 72 hours). The X-axis represents 0.1 μM (–7.0, concentration LogM) to 100 μM (–4.0, concentration LogM). Analysis of Apop in (**h**) MM425X and (**i**) MM390X cells treated with vehicle or niraparib (6 μM). Apop, apoptosis; PARPi, PARP inhibitor; PDX, patient-derived xenograft; qRT-PCR, quantitative real-time reverse transcriptase–PCR; RNA-Seq, RNA sequencing; TG, triglyceride.

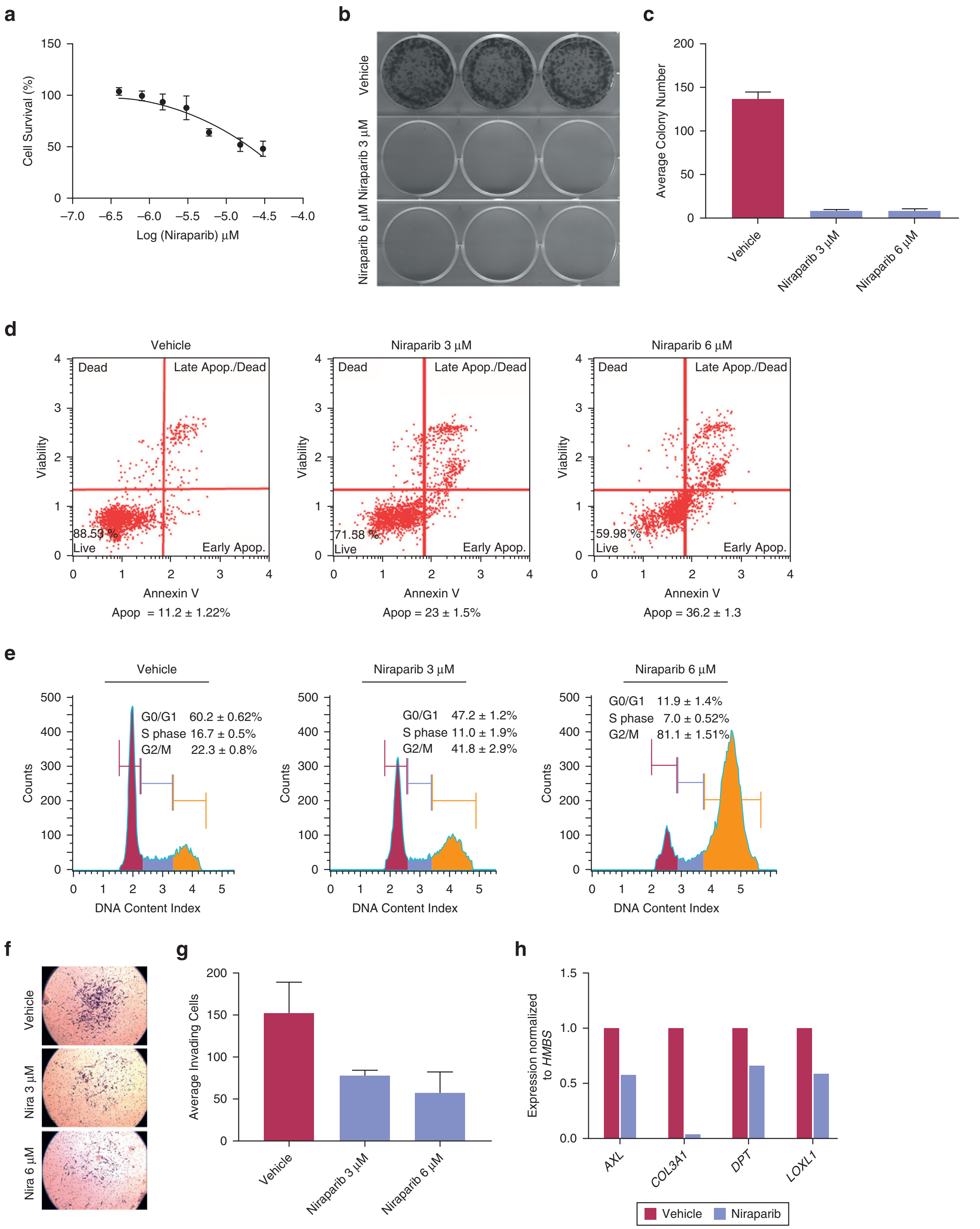

Figure 2. Niraparib regulates viability, Apop, colony formation, cell-cycle progression, and invasion in LOX cells. (**a**) Cell survival analysis of LOX cells treated with niraparib. (**b**) Colony formation ability of LOX cells treated with vehicle or niraparib (3 µM and 6 µM). (**c**) Quantification of colony formation in LOX cells treated as described in **b**. (**d**) Analysis of Apop in LOX cells treated with vehicle or niraparib (3 µM and 6 µM). (**e**) Cell-cycle analysis of LOX cells treated with vehicle or niraparib (3 µM and 6 µM). (**f**) Representative pictograph showing LOX cells invading into Matrigel after niraparib treatment for 48 hours. (**g**) Quantification of LOX cell invasion after niraparib treatment. (**h**) Relative expression of *AXL*, *COL3A1*, *DPT*, and *LOXL1* genes by qRT-PCR in LOX cells treated with vehicle or niraparib (6 µM) for 48 hours and normalized to the human *HMBS* gene. Apop, apoptosis; qRT-PCR, quantitative real-time reverse transcriptase–PCR.

CHD2 mutations as a potential candidate for PARPi therapy, even though CHD2 is not a part of the known HR complex. Furthermore, culture assays indicated that niraparib treatment induced apoptosis and inhibited the viability, colony formation, and invasiveness of melanoma cells. Molecular profiling of treated PDX tumor tissues identified a unique gene expression signature of response to niraparib, characterized by inhibition of cell motility and matrix remodeling, which represent the hallmarks of invasive metastatic melanoma. Specifically, we show that niraparib treatment inhibited the expression of collagen genes (*COL3A1* and *COL1A2*) in a melanoma PDX. In addition to its role in extracellular matrix regulation, COL3A1 expression level is a negative prognostic biomarker in ovarian carcinoma (Engqvist et al., 2019). *LOX* levels have been associated with aggressive tumor growth and shorter survival in uveal melanoma (Abourbih et al., 2010). This enzyme and the related protein LOXL1 directly regulate the cross linking of extracellular matrix proteins and thus regulate cancer metastasis (Xiao and Ge, 2012), including promoting melanoma lung metastasis (Lee et al., 2011). Importantly, niraparib treatment of melanoma PDX samples inhibited the expression levels of AXL. Elevated AXL expression has recently been shown to be associated with therapeutic resistance in melanoma and correlated with an invasive gene signature (Konieczkowski et al., 2014; Müller et al., 2014). A few of the niraparib-upregulated genes (*ALDOC* and *FADS2*) are associated with treatment-induced tumor metabolic reprogramming (Ward and Thompson, 2012). Taken together, our results suggest that niraparib treatment disrupts multiple key pathways important to tumor progression, including some that have not been previously described for any PARPi.

Our preclinical results are consistent with emerging evidence in other cancer types that have identified HR deficiency as a useful predictive biomarker for PARPi treatment. In two randomized trials in patients with germline *BRCA*-mutant, HER-2–negative advanced breast cancer, PARPi treatment was associated with significant progression-free survival improvement over physician's choice of chemotherapy, for olaparib over chemotherapy and for talazoparib over chemotherapy (Litton et al., 2018; Robson et al., 2017). Similar benefits in progression-free survival have been shown with various PARPis in patients with metastatic ovarian cancer with either a germline or somatic *BRCA* mutation (Swisher et al., 2017), in patients with germline *BRCA*-mutant pancreatic cancer (Golan et al., 2019), and most recently, in patients with castration-resistant prostate cancer (de Bono et al., 2020). Intriguingly, PARPis also impart a clinical benefit to patients with nongermline *BRCA*-mutant, HR-deficient malignancies. In a phase III study of niraparib in patients with recurrent ovarian cancer, maintenance therapy with niraparib had a statistically significant survival advantage over placebo, including a clinically meaningful benefit to PARP inhibition in patients with cancer with HR deficiency, regardless of *BRCA* mutation status (Mirza et al., 2016). Similar findings were reported with olaparib and rucaparib, another PARPi, in prostate cancer (Abida et al., 2020; de Bono et al., 2020) and rucaparib in ovarian cancer (Swisher et al., 2017). Whereas two clinical trials evaluated the clinical efficacy of PARP inhibition in combination with temozolomide in patients with metastatic melanoma (Middleton et al., 2015; Plummer et al., 2013), these studies were conducted in unselected patient populations. Thus, to date, the antitumor activity of PARP inhibition in melanomas with HR-DDR gene alteration has not been evaluated. It will be important to determine whether (and which of) these HR-DDR pathway gene mutations, which have been reported to cause significant disruption in DNA repair, will result in sensitivity to PARPi in patients with advanced melanoma. On the basis of our finding of frequent HR deficiency and the strong rationale for a PARPi in this patient population, we have initiated a phase II clinical trial of niraparib in patients with advanced melanoma harboring an HR-DDR pathway gene mutation and are currently enrolling patients (NCT03925350).

In summary, our study characterized the HR-DDR–mutant cohort in cutaneous melanoma, supported by analysis of two large, molecularly profiled cohorts of patients with melanoma, and identified previously unreported mutations in various HR-DDR pathway genes in melanoma. However, analyses of additional data sets will be necessary to confirm the significant association observed between HR-DDR pathway gene mutations and response to checkpoint inhibitor therapy. In addition, our preclinical studies suggest robust activity of PARPi in the setting of HR deficiency in melanoma and have identified a unique gene signature of response to PARP inhibition. Thus, our results suggest that melanomas with these alterations may define a unique subset of patients with distinct biologic and clinical characteristics. More importantly, our results suggest the potential clinical efficacy of PARP inhibition in patients with melanoma with HR pathway gene mutations, a hypothesis that is currently being tested in a clinical trial.

MATERIALS AND METHODS

Approval for this study, including a waiver of informed consent and a Health Insurance Portability and Accountability Act waiver authorization, was obtained from the Sutter Health Institutional Review Board and the Western Institutional Review Board. We searched the Institutional Tumor Registry Database and Melanoma Center Database at California Pacific Medical Center (San Francisco, CA) for patients with a diagnosis of metastatic melanoma. We performed a retrospective medical record review of all patients with metastatic cutaneous melanoma whose tumors were analyzed by the hybrid capture–based next-generation sequencing technique (FoundationOne) that included the entire coding sequence of 315 cancer-related genes performed by FMI using formalin-fixed, paraffin-embedded tumor samples (Frampton et al., 2013). For this analysis, we defined the following genes as HR-DDR pathway genes: *ARID1A, ARID1B, ATM, ATR, ATRX, BAP1, BARD1, BLM, BRCA1, BRCA2, BRIP1, CHEK2, FANCA/C/D2/E/F/G/L, MRE11, NBN, PALB2, RAD50, RAD51, RAD51B/C,* and *WRN*. Mutations were included on the basis of information contained in the Catalog of Somatic Mutations in Cancer and ClinVar databases for evidence of pathogenicity of the specific alteration. From the sequencing analysis, TMB was determined on 1.1 megabase pair of sequenced DNA and calculated as previously described (Chalmers et al., 2017). High and low TMBs were defined as >20 and <6 mutations per megabase, respectively. The patient and tumor characteristics were obtained from electronic health records. We included patients with advanced melanoma of cutaneous origin and of unknown primary cutaneous site and excluded mucosal and uveal melanomas in this analysis. In addition, we included the frequency of HR pathway gene mutations among 1,986 patients with melanoma whose tumors were assessed by FoundationOne at Foundation Medicine (Supplementary Table S3).

Treatment studies in vivo

All experiments were conducted under a California Pacific Medical Center Institutional Animal Care and Use Committee–approved protocol (protocol number 18.03.03; Institutional Animal Care and Use Committee assurance number D16-00015). Three cohorts of female NSG mice aged 8 weeks (strain NOD.Cg-Prkdc^{scid} IL2rg^{tm1Wjl}/SzJ from Jackson Laboratories, Sacramento, CA) bearing subcutaneous metastatic melanoma PDX tumors were randomized according to initial tumor volume. Mice were treated with either vehicle (control group), 25 mg/kg niraparib (, or 50 mg/kg olaparib. Tumors were measured approximately once weekly by an operator blinded to the treatment groups. Tumor volume was calculated as (length $\times$ width^2)/2.

RNA sequencing and bioinformatics data analysis

RNA extraction from flash-frozen tissue samples was performed as previously described (Olsson et al., 2016; Salomonis et al., 2010; Soroceanu et al., 2013). Total RNA was extracted from PDX tumor tissues using the RNeasy tissue kit (Qiagen, Hilden, Germany) after tissue disruption using ruptor disposable probes, and the concentration was measured by Qubit RNA HS Assay Kit (Thermo Fisher Scientific, Waltham, MA). RNA sequencing was performed from ∼500 ng of total RNA processed using TruSeq polyA selection at a target depth of 40 million paired-end, stranded reads on an Illumina 2500 (Illumina, San Diego, CA).

Gene expression analyses

The RNA sequencing data were aligned to the human reference genome (hg19) using the software STAR, followed by gene quantification in the software AltAnalyze to obtain gene-level reads per kilobase of transcript per million mapped read values. Gene expression quantification (Reads Per Kilobase of transcript per Million Mapped Read) was determined from exon–exon junctions, and differential expression (fold >1.2, empirical Bayes moderated *t*-test $P < 0.05$) was performed in AltAnalyze, version 2.1.3, using the Ensembl 72 human database. Embedded gene set enrichment analyses were performed using GO-Elite with default options. Hierarchical clustering was performed in AltAnalyze using Hierarchical Ordered Partitioning and Collapsing Hybrid clustering for rows and weighted cosine clustering for genes. The analyses presented focused on a heatmap representing the top 268 upregulated and downregulated genes (empirical Bayes moderated *t*-test, $P < 0.05$; fold change = 1.2).

Statistical analysis

Prognostic factors were coded in terms of the indexes proposed by the American Joint Committee on Cancer Melanoma Expert Panel (Gershenwald et al., 2017). Thus, primary tumor thickness was classified using the four-point scale defined by the T categories, and age was coded in deciles. The significance of differences in the frequency of various factors between different groups (e.g., HR altered vs. nonaltered) was assessed using chi-square analysis, Fisher exact test, or the *t*-test, as appropriate. The association between the presence of a given factor and OS was assessed using univariate and multivariate Cox regression analysis. All *P*-values reported are two-sided. Additional information on methods is provided in the Supplementary Materials and Methods.

Data availability statement

Datasets related to this article can be found at https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE163424, hosted at National Center for Biotechnology Information Gene Expression Omnibus (Kim et al., 2020).

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CONFLICTS OF INTEREST

KBK has received an honorarium from Foundation Medicine for serving as a consultant, SZM is an employee of Foundation Medicine, and JR is a board member for Foundation Medicine. The remaining authors state no conflict of interest.

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AUTHOR CONTRIBUTIONS

Conceptualization: KBK, LS, DDS, MKS; Data Curation: KBK, LS, DDS, SZM, MN, MKS; Formal Analysis: KC, JRM, NS, AB; Funding Acquisition: DRM, MKS; Investigation: KBK, LS, DDS, SZM, JR, EV, AAD, PYD, RI, MKS; Methodology: KBK, LS, DDS, JRM, NS, MKS; Project Administration: KBK, LS, MKS; Resources: KBK, LS, DDS, SZM, JR, JM, SPL, MIS, BMP, DRM, SM, MKS; Software: LS, DDS, AAD, MN, PYD, RI, KC, JRM, NS, AB; Supervision: KBK, MKS; Validation: KBK, LS, DDS, MKS; Visualization: LS, DDS, AAD, PYD, RI, MKS; Writing – Original Draft Preparation: KBK, LS, DDS, SZM, MKS; Writing – Review and Editing: KBK, LS, DDS, JR, MKS

SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at www.jidonline.org, and at https://doi.org/10.1016/j.jid.2021.01.024.

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SUPPLEMENTARY MATERIALS AND METHODS

Colony formation assay

LOX cells (a generous gift from Dr Oystein Fodstad) were grown in RPMI 1640 media supplemented with 5% fetal bovine serum and 1% penicillin and streptomycin at 37 °C. The colony formation assay was performed as previously described (de Semir et al., 2020). LOX cells (300–500) were plated in each well of a six-well plate and treated with the respective drugs the next day. Cells were allowed to grow till visible colonies appeared, usually 6–10 days after treatment with niraparib (3 μM and 6 μM). Colonies were stained with crystal violet (Sigma-Aldrich, St. Louis, MO) and counted. Data are presented as a bar graph of the average number of colonies along with the pictograph of colonies.

Cell viability assay

Cell viability was assessed as described (de Semir et al., 2020). Briefly, LOX cells (1,000–2,000) were plated in a 96-well plate and treated with niraparib (maximum concentration = 6 μM, six dilutions) the next day. Cell viability was assessed using the Cell Counting Kit-8 (Dojindo Molecular Technologies, Rockville, MD) after 72 hours of drug treatment following the manufacturer's instructions. In total, 20 μl of Cell Counting Kit-8 was added to each well, and the plate was incubated at 37 °C for 2 hours. Absorbance was read at 450 nm.

Cell culture and cell viability assays for metastatic melanoma patient-derived xenograft cells

MM390, MM425, and MM507 patient-derived xenograft (PDX) cultures were grown from PDX tissue, as previously described (Ice et al., 2020). All PDX-derived cells were authenticated using short tandem repeat analysis (ATCC, Manassas, VA; https://www.atcc.org/en/Services/Testing_Services/Cell_Authentication_Testing_Service.aspx); the latest test was performed in 2019 using first and second in vivo passages; cells were also confirmed to be mycoplasma free using a PCR testing service. All experiments described in this manuscript were performed using first or second in vivo passages of metastatic melanoma PDX samples. The cells were kept in culture for the duration of the viability experiments, and fresh batches of cells were used, if needed for repeat experiments.

Master stocks of niraparib (#S2741, Selleck Chemicals, Houston, TX) were made at maximum solubility (62.5 mg/ml or 195 mM) with DMSO. Cell viability was assayed with PDX cells in a 96-well plate (#BD353377, BD Falcon, Franklin Lakes, NJ) at a density of 2,000–3,000 cells per 100 μl of media and/or well in triplicate. Cells were treated the next day with niraparib at different concentrations, and controls were treated with DMSO. Cells were incubated for 72 hours at 37 °C and 5% carbon dioxide. After incubation, a total of 60 μl of CellTiter-Glo 3D Cell Viability (#G9683, Promega, Madison, WI) was added to each well, and cell viability was quantified as per the manufacturer's instruction on luminescence.

Cell-cycle and apoptosis assays

Cell cycle and apoptosis were measured using the Muse cell cycle kit, Muse Annexin V apoptosis kit, and Muse Caspase 3/7 kit (EMD Millipore, Hayward, CA) following the manufacturer's instructions. Cells were plated and treated the next day with the various drugs for 72 hours. For the cell-cycle assay, cells were trypsinized, washed with PBS, fixed with cold ethanol, and incubated at −20 °C for 2 hours. Cells were centrifuged and washed with PBS, followed by the addition of 200 μl of cell-cycle reagent. Cells were incubated for 30 minutes in the dark at room temperature and analyzed by Muse Cell Analyzer. For the apoptosis assay, cells were trypsinized and washed with PBS. Cells were resuspended in 100 μl of apoptosis reagent and 100 μl of 5% RPMI-1640 media, incubated for 20 minutes at room temperature, and analyzed by Muse Cell Analyzer.

Cell invasion assays

LOX cells (75,000 cells) were plated on top of Matrigel-coated transwells in the presence of DMSO or niraparib (3 μM and 6 μM) dissolved in DMEM. Fetal bovine serum (20%) was used as a chemoattractant in the lower chamber. Forty-eight hours later, the filters were cleaned, fixed in 2.5% glutaraldehyde, and stained with Toluidine blue (Sigma-Aldrich). All conditions were run in at least triplicate, and cells were counted for the entire filter area.

In vivo treatment of tumor-bearing mice with PARP inhibitors

Dosing ranges were based on previously published studies, and intraperitoneal dosing was chosen as the standardized route of administration and vehicle formulation between all drugs. Niraparib was diluted in the vehicle formulation: 4% DMSO, 4% Tween-80, and 92% saline. Olaparib was diluted in the vehicle formulation: 10% DMSO, 10% Tween-80, and 80% saline. PDX tumor volumes were measured once per week with calipers, and treatment groups were blinded to measuring personnel. Treatment began when PDX tumor volumes reached approximately 100 mm^3. PDX-bearing NSG mice were intraperitoneally treated daily for 5 consecutive days for approximately 4 weeks for a total of 20 treatments. A total of 8 mice per treatment group for a total of 24 mice were used for each cohort. At the completion of in vivo experiments, all mice were humanely killed using carbon dioxide overdose followed by cervical dislocation. PDX tumor tissues were harvested for RNA analysis by flash freezing and stored at −80 °C until further analysis.

Validation of gene expression using quantitative real-time reverse transcriptase–PCR

Gene expression was assessed using quantitative real-time reverse transcriptase–PCR (TaqMan assay) as previously reported (de Semir et al., 2020). mRNAs were assayed using the TaqMan Gene Expression Assays according to the manufacturer's instructions (Thermo Fisher Scientific, Waltham, MA). The following TaqMan probes were used: *AXL* (Hs01064444_m1), *COL1A2* (Hs01028956_m1), *COL3A1* (Hs00943809_m1), *DPT* (Hs00355056_m1), *LOXL1* (Hs00935937_m1), *ALDH3B2* (Hs02511514_s1), *ALDOC* (Hs00193059_m1), *FADS2* (Hs00188654_m1), and *GSTT2* (Hs00168315_m1). The expression levels of these genes were normalized to the human *HMBS* (Hs00609297_m1) gene before comparisons.

Supplementary Table S1. The 15 Mutations in HR-Deficient Genes from the CPMC Cohort

Gene	Mutation
ATM	L167*
FANCA	3828+1G>C splice
CHEK2	T367fs*15 and 320-1G>A splice
ARID1A	Q1537*; Q575* and G133fs*57
BRCA1	4185+1G>A splice; R1495M and 671-1G>A splice
BAP1	M1I
BRIP1	R798*
ATR	V327fs*3
ATRX	L664fs*1
BARD1	Q666*
PALB2	Q1020*

Abbreviations: CPMC, California Pacific Medical Center; HR, homologous recombination.

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